Math 242, S15 **Name:** Answer

Quiz 6 **SSN: xxx-xx-**

<u>**Instructions:**</u> **There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (10 points) Suppose that a motorboat is moving at 50 ft/s when its motor suddenly quits at time $t = 0$, and that 10 s later the boat has slowed to 25 ft/s. Assume that the resistance it encounters while coasting is proportional to its velocity, i.e., $\frac{dv}{dt} = -kv$.

(a) Find the velocity $v(t)$ of the motorboat at time t.

$$\begin{cases} \frac{dV}{dt} = -kV \\ V(0) = 50 \end{cases}$$

$$\frac{1}{V} dV = -k dt$$

$$\ln|V| = -kt + C$$

$$\Rightarrow |V| = e^{-kt+C}$$

$$V(t) = Ae^{-kt}$$

$$V(0) = 50 \Rightarrow 50 = Ae^0$$

$$A = 50$$

$$\Rightarrow V(t) = 50e^{-kt}$$

In addition

$$V(10) = 25 \Rightarrow 25 = 50e^{-10k}$$

$$e^{-10k} = \frac{1}{2}$$

$$k = \left(-\frac{1}{10}\right)\ln\left(\frac{1}{2}\right) = \frac{\ln 2}{10} \approx 0.0693$$

Thus

$$V(t) = 50e^{-\frac{\ln 2}{10}t}$$

(b) Find the distance $x(t)$ travelled by the motorboat at time t, and how far will the motorboat coast in all?

$$X(t) = \int_0^t V(s)ds = \int_0^t 50e^{-\frac{\ln 2}{10}s} ds$$

$$= \left[-\frac{500}{\ln 2} e^{-\frac{\ln 2}{10}s}\right]_{s=0}^{s=t}$$

$$= \frac{500}{\ln 2}\left(1 - e^{-\frac{\ln 2}{10}t}\right)$$

$\lim_{t\to+\infty} V(t) = 0$ and $V(t) > 0$ for $t > 0$.

Thus the distance the boat coast in all is

$$\lim_{t\to+\infty} X(t) = \lim_{t\to+\infty} \frac{500}{\ln 2}\left(1 - e^{-\frac{\ln 2}{10}t}\right) = \frac{500}{\ln 2} \approx 721.5 \text{ ft.}$$